

Algorithm 2

Energy Histogram

Speech and silence segmentation

Frame 25 ms. Hop 10 ms.

Minimum silence duration: 200 ms

Name / Student ID: pending

Algorithm 2 workflow

1. Read WAV with 25/10 ms framing



2. Compute Energy as mean squared amplitude



3. Build Energy histogram and compare $W=1..50$



4. Hysteresis: merge $<200\text{ms}$ gaps, discard $<100\text{ms}$ speech

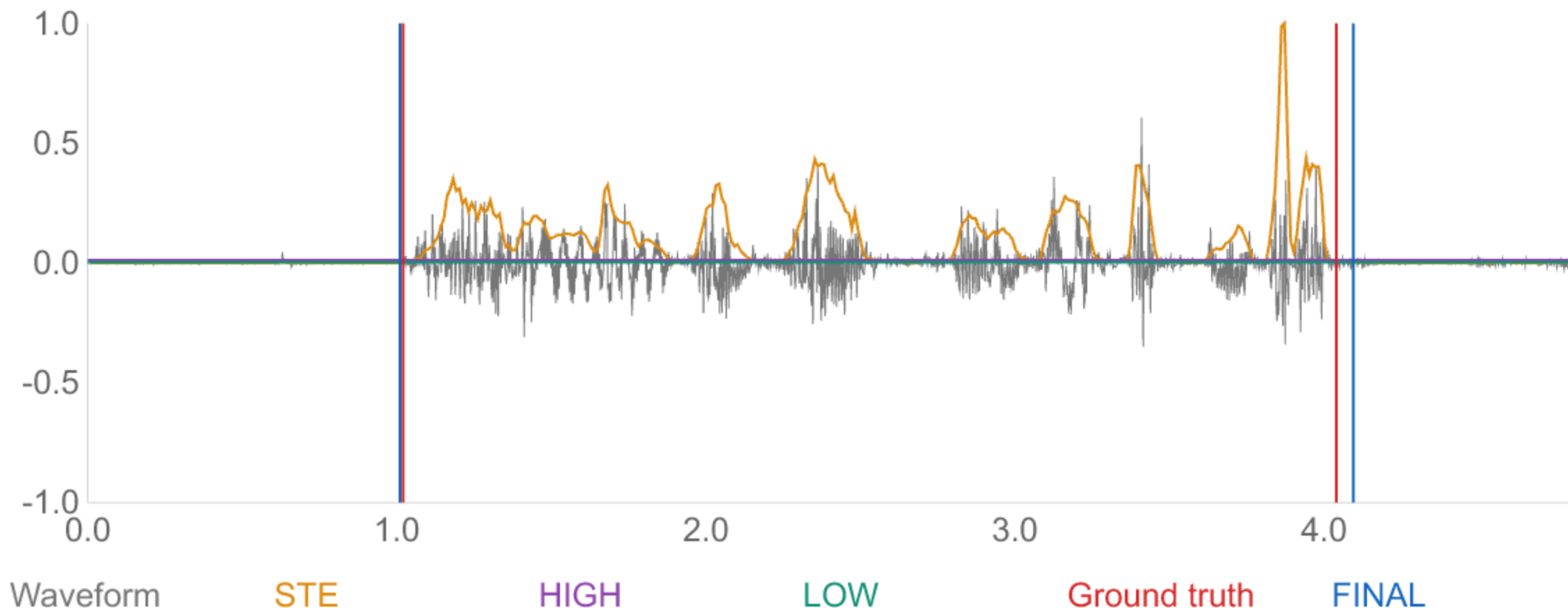


5. Select one W using four test LABs

Train $W=1$. Global $W=20$. All $W=1..50$ tie.

phone_F2: test results

MAE 32.50 ms RMSE 39.53 ms

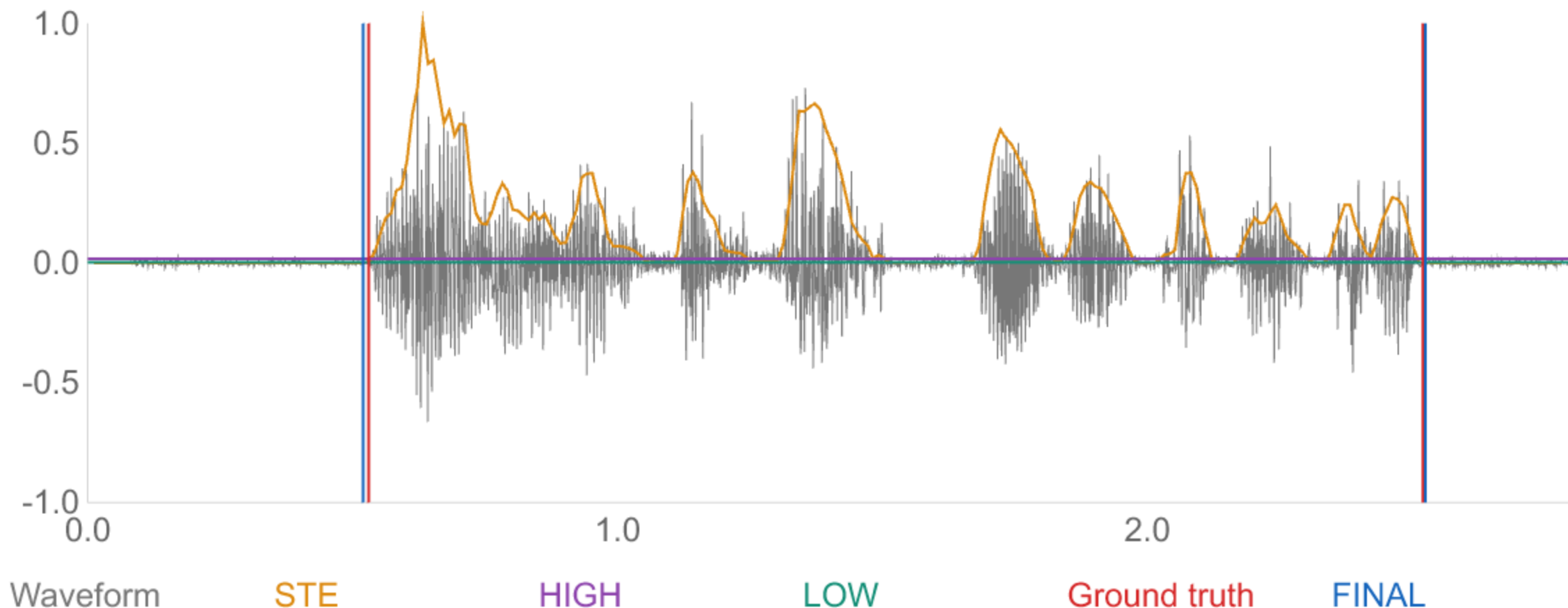


Time (s)

Test LABs tune W. MAE is not independent.

phone_M2: test results

MAE 7.50 ms RMSE 7.91 ms

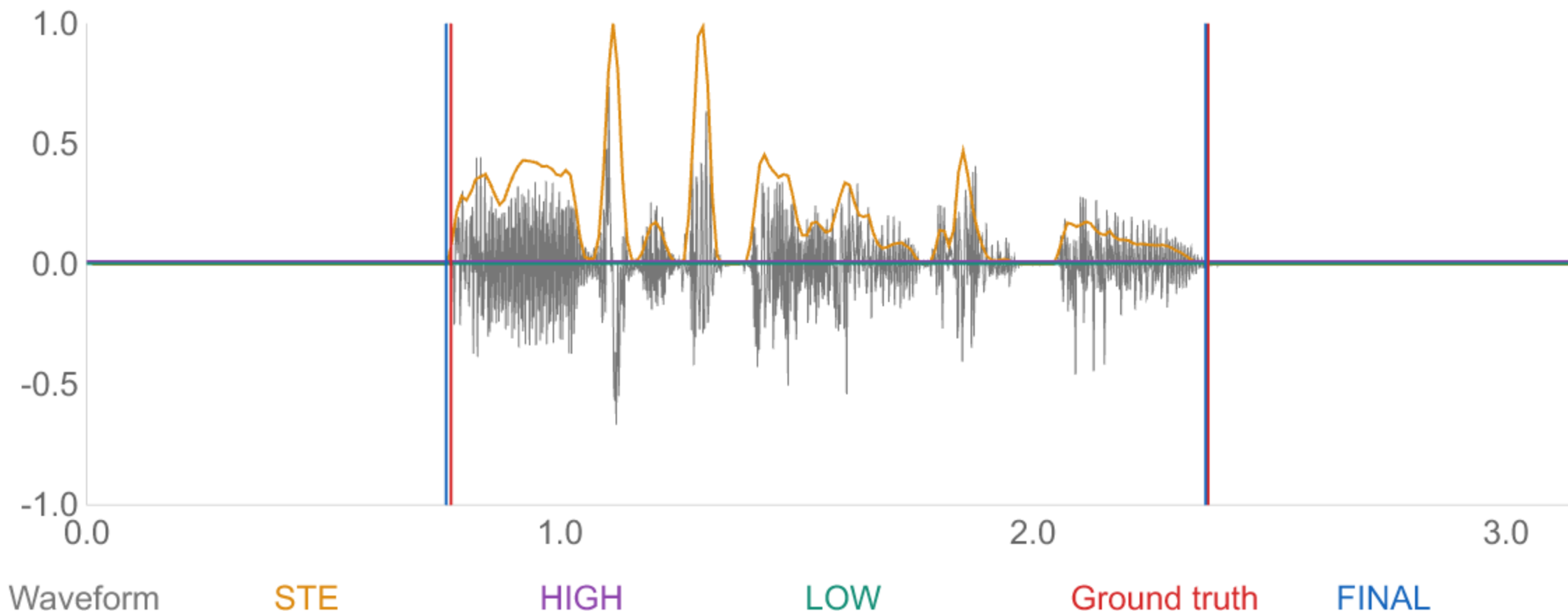


Time (s)

Test LABs tune W. MAE is not independent.

studio_F2: test results

MAE 7.51 ms RMSE 7.91 ms

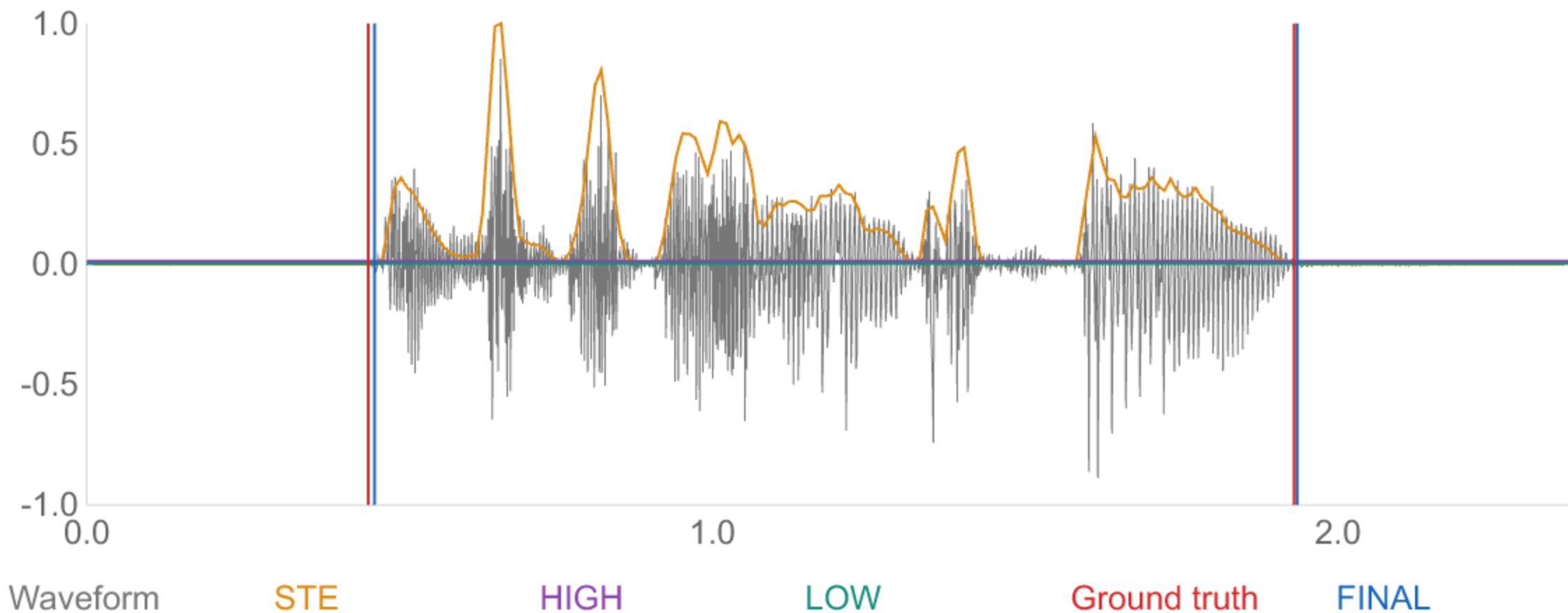


Time (s)

Test LABs tune W. MAE is not independent.

studio_M2: test results

MAE 7.49 ms RMSE 7.90 ms



Time (s)

Test LABs tune W. MAE is not independent.

Error on four test files

File	MAE (ms)	RMSE (ms)
phone_F2	32.50	39.53
phone_M2	7.50	7.91
studio_F2	7.51	7.91
studio_M2	7.49	7.90

Mean MAE: 13.75 ms

Residual noise affects phone_F2.

Test LABs tune W. MAE is not independent.